

Real-Time ASL Recognition System

Project Overview

This project is a real-time American Sign Language (ASL) alphabet recognition system built using TensorFlow, OpenCV, Flask, and MobileNetV2. The system captures webcam input, processes a fixed Region of Interest (ROI), predicts ASL hand signs, and converts them into text in real time.

Objective

The objective of this project is to provide a lightweight and responsive ASL recognition solution without relying on external hand-tracking frameworks such as MediaPipe. The system focuses on efficient real-time inference and a clean user experience.

Tech Stack

- Python
- TensorFlow / Keras
- MobileNetV2
- OpenCV
- Flask
- HTML / CSS / JavaScript

Key Features

- Real-time ASL alphabet recognition
- Fixed ROI webcam processing
- Prediction smoothing using deque
- Sentence formation logic
- Word recommendation system
- Text-to-Speech functionality
- Interactive frontend UI with progress bar
- Manual controls (Backspace, Space, Clear)

System Workflow

Webcam Input → ROI Extraction → Image Preprocessing → MobileNetV2 Model → Prediction Smoothing → Sentence Formation → Flask Frontend UI

Challenges Solved

- Reduced unstable predictions using frame smoothing
- Improved usability with capture progress feedback
- Maintained real-time performance without MediaPipe
- Synchronized backend predictions with frontend updates

Future Improvements

- Dynamic hand detection
- Sentence-level ASL recognition
- Transformer-based sequence modeling
- Mobile deployment support
- Improved dataset diversity

Conclusion

This project demonstrates a practical real-time ASL recognition pipeline integrating computer vision, deep learning, and web technologies into a usable end-to-end application.